Summary in Graph

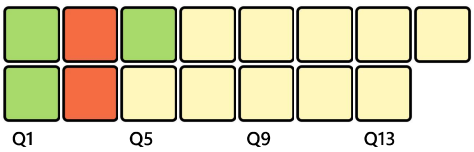
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 2)

Qs. Attempted:	**5** 5 + 0	**Correct Marks:**	**3** 3 + 0
Correct Attempts:	**3** 3 + 0	**Penalty Marks:**	**0.67** 0.67 + 0
Incorrect Attempts:	**2** 2 + 0	**Resultant Marks:**	**2.33** 2.33 + 0

Total Questions:	**15** 10 + 5
Total Marks:	**20** 10 + 10
Exam Duration:	**45 Minutes**
Time Taken:	**45 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (51) out of 512 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider a dataset with two features and three points: $x_q = (0, 0)$ is the query point, and the other points are $x_1 = (1, 1)$ and $x_2 = (1.5, 0)$. The task is to determine the 1-Nearest-Neighbor (1-NN) for the query point x_q using two different distance metrics: Euclidean and Manhattan distances.
Which of the following statements is true about the nearest neighbor of the query point x_q based on the distance metric used?

A. x_1 is the nearest neighbor according to both Euclidean and Manhattan distances.

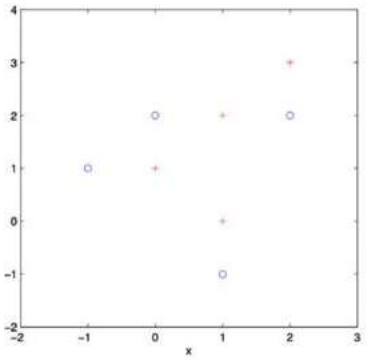
B. x_2 is the nearest neighbor according to both Euclidean and Manhattan distances.
C. x_1 is the nearest neighbor according to Euclidean distance, and x_2 is the nearest neighbor according to Manhattan distance.
D. x_2 is the nearest neighbor according to Euclidean distance, and x_1 is the nearest neighbor according to Manhattan distance.

Your Answer: C | Correct Answer: C | Correct | Time taken: 11min 19sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following training set in a two-dimensional feature space:

(x , y)	Class
(-1 , 1)	-
(0 , 1)	+
(0, 2)	-
(1 , -1)	-
(1 , 0)	+
(1 , 2)	+
(2 , 2)	-
(2 , 3)	+



What are the predictions of the k-nearest-neighbor classifier at the point $(1, 1)$ for different values of k ?

A. 3-nearest: +, 5-nearest: +, 7-nearest: -
B. 3-nearest: -, 5-nearest: -, 7-nearest: -
C. 3-nearest: +, 5-nearest: -, 7-nearest: +
D. 3-nearest: -, 5-nearest: +, 7-nearest: +

Your Answer: A | Correct Answer: A | Correct | Time taken: 02min 03sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

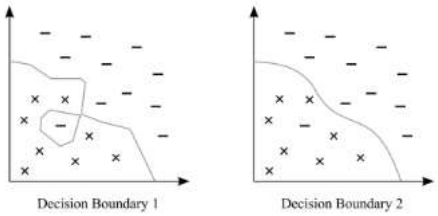
Imagine you are using a k-Nearest Neighbor classifier on a dataset with lots of noise. You want your classifier to be less sensitive to the noise. Which of the following is likely to help and with what side effect?

A. Increase the value of $k \rightarrow$ Increase in prediction time
B. Decrease the value of $k \rightarrow$ Increase in prediction time
C. Increase the value of $k \rightarrow$ Decrease in prediction time
D. Decrease the value of $k \rightarrow$ Decrease in prediction time

Your Answer: D | Correct Answer: A | Incorrect | Time taken: 03min 35sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Figure shows the decision boundaries (DB) for two nearest-neighbour classifiers - 1NN and 3NN. Consider following statements and choose the correct option (True/False for all the statements 1/2/3/4).



1. DB1 belongs to $3NN$ while DB2 belongs to $1NN$.
2. DB1 belongs to $1NN$ while DB2 belongs to $3NN$.
3. DB1 gives zero test error.
4. DB1 gives zero training error.

A. 1: True, 2: False, 3: True, 4: False
B. 1: False, 2: True, 3: False, 4: True
C. 1: False, 2: True, 3: True, 4: False
D. 1: True, 2: False, 3: False, 4: True

Your Answer: D | Correct Answer: B | Incorrect | Time taken: 12min 22sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You are building a k-NN classifier to filter spam emails. You divide 100 labeled emails into a training set (60 emails), a validation set (20 emails), and a test set (20 emails). For $k = 1$, you get a training error of 0.0% and a validation error of 11.2%. For $k = 2$, you get a training error of 8.5% and a validation error of 9.4%. Which classifier should you expect to have a lower error on the test set, and why?

A. The $k = 1$ classifier, because it has a 0.0% training error.
B. The $k = 2$ classifier, because it has a lower validation error.
C. The $k = 1$ classifier, because it will generalize better with less training error.
D. Both classifiers are equally likely to have the same error on the test set.

Your Answer: B | Correct Answer: B | Correct | Time taken: 01min 06sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You are using a 1-NN classifier to filter spam emails. Given a training set of size n and a validation set of size m, and assuming that the distance calculation takes constant time, what is the asymptotic time complexity of computing the validation error? (Use big-O notation in terms of n and m.)

A. $O(n)$
B. $O(m)$
C. $O(n + m)$
D. $O(n \cdot m)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 14min 53sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

The table below shows the test set for a **1-nearest-neighbor classifier** that uses Manhattan distance, i.e., the distance between two points at coordinates p and q is $|p-q|$. The only attribute, X, is real-valued, and the label, Y, has two classes, 0 and 1 . Suppose a subset containing $n \leq 8$ examples is selected from this set to train the classifier, and the accuracy of the classifier is 100 percent when tested on this set (with all 8 examples). What is the smallest possible value for n ? In case of ties in distance, use the example with smallest X value as the neighbor.

X	-5	-4	-1	0	1	3	4	8
Y	0	1	0	0	0	0	0	1

A. 2
B. 3
C. 4
D. 5

Your Answer: **Correct Answer: C** Not Attempted Time taken: 00min 00sec Discuss

Q #8 Multiple Select Type Award: 1 Penalty: 0 Machine Learning

Which of the following statements about k-Nearest Neighbor (k-NN) are true in a classification setting, and for all k ? Select all that apply.

A. The training error of a 1-NN will always be better than or equal to that of $5-\mathrm{NN}$.
B. The test error of a 1-NN will always be better than that of a 5-NN.
C. The decision boundary of the k-NN classifier is linear.
D. The time needed to classify a test example with the k-NN classifier grows with the size of the training set.

Your Answer: **Correct Answer: A;D** Not Attempted Time taken: 00min 00sec Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Machine Learning

Which of the following is true about the k-nearest neighbors (KNN) algorithm?

A. It is a parametric model.
B. It can learn a nonlinear decision boundary between classes.
C. It requires a separate training phase and testing phase for prediction.
D. It typically requires longer training compared to other ML algorithms.

Your Answer: **Correct Answer: B** Not Attempted Time taken: 00min 00sec Discuss

Q #10 Multiple Choice Type Award: 1 Penalty: 0.33 Machine Learning

In k-nearest neighbor (kNN) regression, the prediction of Y at point x_0 is given by the average of the values Y at the k neighbors closest to x_0. Denote the ℓ-nearest neighbor to x_0 by $x_{(\ell)}$ and its corresponding Y value by $y_{(\ell)}$. Which of the following expresses the prediction $\hat{f}\left(x_0\right)$ in terms of $y_{(\ell)}$, for $1 \leq \ell \leq k$?

A. $\hat{f}\left(x_0\right)=\frac{1}{k} \sum_{\ell=1}^k y_{(\ell)}$
B. $\hat{f}\left(x_0\right)=\frac{1}{k} \sum_{\ell=1}^k\left(y_{(\ell)} \cdot e^{-\left|x_{(\ell)}-x_0\right|}\right)$
C. $\hat{f}\left(x_0\right)=\frac{1}{k} \sum_{\ell=1}^k\left(y_{(\ell)}+\left(x_{(\ell)}-x_0\right)^2\right)$

D. $\hat{f}\left(x_0\right)=\frac{1}{k} \sum_{\ell=1}^k\left(y_{(\ell)}-\left(x_{(\ell)}-x_0\right)\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Let S_1 and S_2 be two separate training data sets. Let $h_k\left(S_i, x\right)$ be the binary classifier function which takes in a training data set S_i and a test example x, and outputs the result of the k-nearest neighbor classifier that has been trained on data set S_i on the test example x. (So, for example, $h_5\left(S_1, x\right)$ is the output of the 5-nearest neighbor classifier trained on S_1 on the test example x). Assume that the two labels are 0 and 1.
Consider the following two statements:
(a) If $h_1\left(S_1, x\right)=1$ and $h_1\left(S_2, x\right)=1$, then $h_1\left(S_1 \cup S_2, x\right)=1$
(b) If $h_3\left(S_1, x\right)=1$ and $h_3\left(S_2, x\right)=1$, then $h_3\left(S_1 \cup S_2, x\right)=1$
Which of the following statements is correct?

A. Only (a) is correct.
B. Only (b) is correct.
C. Both (a) and (b) are correct.
D. Both (a) and (b) are incorrect.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Given the following training set of four labeled points:

$$(-0.8,0),(-0.4,1),(0.2,1),(0.8,0)$$

What is the decision boundary of a 1-NN classifier using this training set?

A.

$$h(x)=\begin{cases}1 & \text{if } -0.4 \leq x \leq 0.8 \\ 0 & \text{otherwise}\end{cases}$$

B.

$$h(x)=\begin{cases}1 & \text{if } -0.8 \leq x \leq 0.2 \\ 0 & \text{otherwise}\end{cases}$$

C.

$$h(x)=\begin{cases}1 & \text{if } -0.6 \leq x \leq 0.5 \\ 0 & \text{if } x<-0.6 \text{ or } x>0.5\end{cases}$$

D.

$$h(x)=\begin{cases}1 & \text{if } 0.2 \leq x \leq 0.8 \\ 0 & \text{otherwise}\end{cases}$$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

A 1-NN classifier will have non-zero training error on a data set with three points (x, y) if:

A. This is not possible for a 1-NN classifier with any data set.
B. Two or more points have the same x-value but different labels.

C. The points are linearly separable.
D. The distance between all points is equal .

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

Which of the following statements is/are TRUE about the k-NN classifier?

A. If point A is among point B's k-nearest neighbors, then point B is always among point A 's k-nearest neighbors.
B. On a training set with 1000 positive items and 1000 negative items, it is possible for 1-NN to have a training set accuracy of 0%, given that there is no overlap in points.
C. Multiplying each dimension of every item's feature vector by 0.1 (assuming Euclidean distance) will not change k-NN results.
D. As the value of k used in a k-NN classifier is incrementally increased from 1 to n , the total number of training examples, the classification accuracy on the training set will always increase.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

You are using a **Probabilistic 3NN** model for binary classification with labels $\{-1, +1\}$. The model predicts the label of a query point based on the relative proportion of labels among its 3 nearest neighbors. For example, if the 3 nearest neighbors have labels $\{-1, -1, +1\}$, the model assigns a probability of $2/3$ to label -1 and $1/3$ to label +1 .
Given the 5 nearest neighbors and their distances from the query point q :

Neighbor	Distance from q	Label
N1	1.2	-1
N2	1.5	+1
N3	1.7	-1
N4	2.0	-1
N5	2.5	+1

What is the predicted probability $P(y = -1)$ using the 3 nearest neighbors?

A. $1/3$
B. $2/3$
C. $3/4$
D. $1/2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff